

Tutorial 7

PS 7.4 PCA for SVD (SVD = singular value decomposition)

$$A = USV^T \quad R^{m \times n}$$

$$U^T U = I$$

$$U \in R^{n \times m}$$

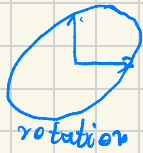
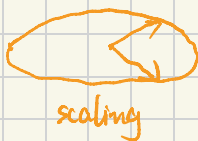
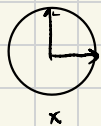
$$V^T V = I$$

$$V \in R^{n \times n}$$

$$S = \text{diag}(s_1, s_2, \dots, s_m) \\ \in R^{m \times m}$$

$$Ax = USV^T x$$

rot
scale
rot



1) mean-subtracted

$$\underline{X} = [\underline{x}_1, \underline{x}_2, \dots, \underline{x}_n]$$

$$\bar{X} = [x_1 - \mu, x_2 - \mu, \dots, x_n - \mu]$$

$$= \begin{bmatrix} \vdots \\ x_1 \\ \vdots \\ x_n \\ \vdots \end{bmatrix} - \begin{bmatrix} \vdots \\ \mu \\ \vdots \\ \mu \\ \vdots \end{bmatrix}$$

$$\mu[1, \dots, 1]^T = \mu \mathbf{1}^T$$

$$\mu = \frac{1}{n} \sum x_i = \frac{1}{n} [x_1, \dots, x_n] \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = \frac{1}{n} \mathbf{1}^T$$

$$\bar{X} = X - \frac{1}{n} \mathbf{1} \mathbf{1}^T \\ = X(I - \frac{1}{n} \mathbf{1} \mathbf{1}^T)$$

$$b) \bar{X} = USV^T$$

$$\bar{Z} = \frac{1}{n} \sum_i (x_i - \mu)(x_i - \mu)^T = \frac{1}{n} \bar{X} \bar{X}^T$$

$$= \frac{1}{n} (USV^T)(USV^T)^T = \frac{1}{n} USV^T V S U^T$$

$$= \frac{1}{n} US^2 U^T = U \begin{bmatrix} s_1^2 & & \\ & \ddots & \\ & & s_n^2 \end{bmatrix} U^T$$

$\therefore (u_i, \frac{1}{n} s_i^2)$ is an eigenvector-value pair of $\bar{Z}$

PCA using SVD

$$1) \bar{X} = X(I - \frac{1}{n} \mathbf{1} \mathbf{1}^T)$$

2) solve SVD of $\bar{X}$

$$3) \frac{1}{n} s_1^2 > \dots > \frac{1}{n} s_n^2 \text{ decreasing order}$$

4) select top k singular value $\Phi = [u_1, \dots, u_k]$

5) PCA co-efficient $Z_k = \Phi^T [X_k - \mu]$

$$\bar{Z} \quad R^{d \times d} \quad \text{eig}(\bar{Z}) = O(d^3)$$

$$\bar{X} \quad R^{d \times n} \quad \text{svd}(\bar{X}) = O(d \cdot n \cdot \min(d, n))$$

$$d < n$$

$$d^2 n > d^3$$

$$d > n$$

$$d n^2 < d^3$$

faster
 $\bar{Z}$
 $\bar{X}$

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PS 7.6 Fisher's Linear discriminant

$$W^* = \arg \max_W \frac{(m_1 - m_2)^2}{s_1 + s_2} \quad m_i = W^T \mu_i \quad s_i = W^T S_i W$$

$$= \arg \max_W \frac{(W^T \mu_1 - W^T \mu_2)^2}{W^T S_1 W + W^T S_2 W} = \frac{[W^T (\mu_1 - \mu_2)]^2}{W^T (S_1 + S_2) W}$$

Within Class Scatter $S_W = S_1 + S_2$ Between Class Scatter $M.B.T.$

$$= \arg \max_W \frac{W^T S_B W}{W^T S_W W}$$

fix the denominator $W^T S_W W = 1$

$$W^* = \arg \max_W W^T S_B W, \text{ s.t. } W^T S_W W = 1$$

Use Lagrange multiplier

$$L(W, \lambda) = W^T S_B W - \lambda (W^T S_W W - 1)$$

$$\frac{\partial L}{\partial W} = 2 S_B W - \lambda 2 S_W W = 0$$

$$S_B W = \lambda S_W W \Leftarrow \text{Generalized eigenvalue problem}$$

$$S_W^{-1} S_B W = \lambda W$$

$$\underbrace{S_W^{-1}}_{\text{vector}} (\underbrace{\mu_1 - \mu_2}_{\text{scalar}}) (\underbrace{\mu_1 - \mu_2}_{\text{scalar}})^T \underbrace{W}_{\text{vector}} = \lambda \underbrace{W}_{\text{vector}}$$

$$W^* \propto S_W^{-1} (\mu_1 - \mu_2)$$

Since the length of w does not matter for the optimisation

$$S_W = S_1 + S_2$$